Difficulty-Calibrated Unlearning Auditing: Exposing Per-Sample Privacy Gaps in Machine Unlearning

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Abstract

Machine unlearning methods are typically evaluated using aggregate membership inference attack (MIA) metrics, which average privacy leakage across all forgotten samples. We show that this practice systematically masks severe per-sample privacy failures: samples that are intrinsically harder to unlearn remain highly vulnerable even when aggregate metrics appear satisfactory. We propose *Difficulty-Calibrated Unlearning Auditing* (DCUA), a diagnostic framework that stratifies forget-set samples into difficulty quintiles using reference-model calibration and evaluates unlearning quality per stratum. Across five unlearning algorithms, three datasets, and three random seeds, we find that the worst-quintile MIA-AUC (WQ-AUC) exceeds the aggregate AUC by 0.08–0.35 absolute, with difficulty gaps (DG) reaching 0.30–0.65. All 15 method–dataset configurations show statistically significant overestimation by aggregate metrics ($p < 0.005$). We also investigate Difficulty-Aware Unlearning (DAU), an adaptive reweighting defense. While DAU shows promising results for Random Labels unlearning (reducing WQ-AUC by up to 0.106 on Purchase-100), it provides negligible improvement for most other methods and can worsen privacy for SCRUB—an informative negative result highlighting that difficulty-aware loss reweighting alone is insufficient to close the per-sample privacy gap. Our findings establish DCUA as a practical, computationally efficient diagnostic that any unlearning evaluation should include, and identify closing the difficulty-dependent privacy gap as an important open problem.

1 Introduction

Machine unlearning—the process of removing the influence of specific training data from a trained model—has become critical for compliance with privacy regulations such as the GDPR's "right to be forgotten" [Bourtoule et al., 2021]. A central challenge is *verification*: determining whether unlearning actually removed the targeted information. The dominant approach uses membership inference attacks (MIAs) as a proxy: if an attacker cannot distinguish whether a sample was in the training set by examining the unlearned model, unlearning is deemed successful [Chen et al., 2021, Kurmanji et al., 2023].

However, standard MIA-based evaluations compute a single aggregate metric (e.g., AUC) over the entire forget set. Recent work has shown this is fundamentally misleading. Hayes et al. [2025] demonstrated that per-example attacks reveal significant privacy failures masked by aggregate metrics, creating a "false sense of privacy." Concurrently, the unlearning difficulty literature has shown that success varies dramatically across individual samples [Rizwan et al., 2024, Zhao et al., 2024, Cheng et al., 2026].

The gap we address. While prior work has diagnosed the problem—that per-sample unlearning quality varies and aggregate metrics are misleading—the question of *how to efficiently expose these failures in practice* remains open. Hayes et al. [2025] require per-example shadow models (computationally expensive), while Zhao et al. [2024] and Rizwan et al. [2024] focus on understanding difficulty factors rather than providing a practical

evaluation protocol. No existing work bridges the difficulty calibration framework from the MIA literature [Watson et al., 2022] into unlearning evaluation.

Our approach. We propose *Difficulty-Calibrated Unlearning Auditing* (DCUA), which uses reference-model difficulty scores [Watson et al., 2022, Carlini et al., 2022] to stratify forget-set samples into difficulty quintiles, then evaluates unlearning quality per stratum. This yields two targeted metrics: *Worst-Quintile AUC* (WQ-AUC), capturing the privacy of the most vulnerable samples, and *Difficulty Gap* (DG), measuring how much privacy protection varies with sample difficulty. We also investigate *Difficulty-Aware Unlearning* (DAU), an adaptive defense that reweights the unlearning loss by sample difficulty.

Our contributions are:

- We propose DCUA, a stratified evaluation framework that uses reference-model difficulty calibration to expose per-sample privacy gaps in machine unlearning. DCUA is computationally efficient—K reference models serve all samples, compared to $O(n)$ shadow models for per-example attacks.
- We demonstrate across 5 unlearning methods, 3 datasets, and 3 seeds that aggregate MIA-AUC systematically overestimates unlearning quality: WQ-AUC exceeds aggregate AUC by 0.08–0.35 absolute in all 15 configurations ($p < 0.005$), with difficulty gaps of 0.26–0.65.
- We investigate DAU as an adaptive defense, finding that simple difficulty-aware loss reweighting is insufficient for most unlearning methods—an important negative result that identifies closing the difficulty-dependent privacy gap as an open problem.

Section 2 reviews related work. Section 3 describes our method. Section 4 presents experiments. Section 5 discusses findings and limitations. Section 6 concludes.

2 Related Work

Machine unlearning evaluation. Hayes et al. [2025] is the closest prior work to our evaluation contribution. They distinguished *population U-MIAs* (one attacker for all examples) from *per-example U-MIAs* (a dedicated attacker per example), showing that aggregate metrics create a false sense of privacy. Our work differs in three ways: (1) we use difficulty calibration [Watson et al., 2022] as a computationally cheaper stratification mechanism, (2) we introduce specific metrics (WQ-AUC, DG) for quantifying the difficulty-dependent gap, and (3) we investigate a defense (DAU). Naderloui et al. [2025] proposed RULI, a per-sample targeted attack framework; our stratified approach provides complementary diagnostic information.

Unlearning difficulty. Rizwan et al. [2024] identified four instance-level factors making unlearning difficult (memorization, atypicality, loss landscape, gradient alignment). Zhao et al. [2024] showed that forget-set characteristics substantially affect unlearning difficulty and proposed RUM, a meta-algorithm that partitions forget sets into homogeneous subsets. Cheng et al. [2026] proposed circuit-level difficulty metrics. Our difficulty estimation via reference models captures a complementary signal specifically designed to connect to MIA vulnerability, and our DCUA framework provides a practical evaluation protocol.

Difficulty calibration in MIA. Watson et al. [2022] established that difficulty calibration dramatically improves MIA accuracy by adjusting membership scores based on sample hardness. Carlini et al. [2022] developed LiRA, which implicitly accounts for difficulty through reference model comparisons. He et al. [2024] identified limitations and proposed RAPID. Our contribution bridges difficulty calibration from the attack domain into the unlearning evaluation domain.

Machine unlearning algorithms. Bourtoule et al. [2021] introduced SISA training for exact unlearning. Kurmanji et al. [2023] proposed SCRUB, a teacher-student distillation framework. These methods are evaluated primarily using aggregate metrics; our work shows these are misleading and proposes stratified alternatives.

Table 1: Comparison with related work. DCUA provides both evaluation and defense investigation, using computationally efficient reference-model calibration.

	Hayes et al.	Rizwan et al.	Zhao et al.	Cheng et al.	**Ours**
Evaluation protocol	✓				✓
Defense proposed			RUM		DAU
Difficulty mechanism	Per-example	4 factors	Partitioning	Circuits	Ref. models
Computational cost	$O(n)$	Analysis	Moderate	Analysis	$O(K)$
MIA connection	U-MIA	None	None	None	Calibration

Per-instance privacy and unlearning. Sepahvand et al. [2025] leveraged per-instance differential privacy bounds for unlearning, focusing on noisy fine-tuning with DP machinery. DAU is algorithm-agnostic and works with any gradient-based unlearning method without requiring DP.

3 Method

3.1 Problem Setting

Consider a model f_θ trained on dataset $D = D_r \cup D_f$, where D_f is the *forget set* (samples to be unlearned) and D_r is the *retain set*. An unlearning algorithm $\mathcal{U}$ produces an unlearned model $f_{\theta'}$ that should behave as if D_f were never in the training data. The gold standard is *retraining from scratch* on D_r alone, producing f_{θ^*}.

Standard evaluation computes the aggregate MIA-AUC: the area under the ROC curve for distinguishing D_f (members) from held-out test samples (non-members) based on the unlearned model's behavior. A lower AUC (closer to 0.5) indicates better unlearning.

3.2 Difficulty-Calibrated Unlearning Auditing (DCUA)

Step 1: Difficulty estimation. We train K reference models $\{f_k\}_{k=1}^K$ on data from the same distribution as D but disjoint from D_f. For each sample $x \in D_f$, we compute the *reference difficulty score*:

$$d(x) = \frac{1}{K} \sum_{k=1}^{K} \mathcal{L}(f_k, x), \tag{1}$$

where $\mathcal{L}$ is the cross-entropy loss. This mirrors the reference-model approach used in LiRA [Carlini et al., 2022] but applied to characterize sample difficulty rather than perform the attack itself. Samples with high $d(x)$ are intrinsically harder—they are atypical or lie in difficult regions of the data manifold.

Step 2: Stratification. We assign each sample to a difficulty quintile Q_1 (easiest, 0–20th percentile) through Q_5 (hardest, 80–100th percentile) based on percentile ranks of $d(x)$. For fair evaluation, we construct *difficulty-matched* non-member sets: for each quintile Q_i, we select test-set samples whose difficulty score falls in the same percentile range, ensuring that within-stratum MIA comparison is not confounded by difficulty.

Step 3: Per-stratum evaluation. For each quintile Q_i, we compute the MIA-AUC using members from Q_i and difficulty-matched non-members. This yields our two key metrics:

- **Worst-Quintile AUC (WQ-AUC)** $= \mathrm{AUC}(Q_5)$: the MIA success rate on the hardest-to-unlearn samples. This is the metric most relevant for privacy, since the hardest samples are the most vulnerable.
- **Difficulty Gap (DG)** $= \mathrm{AUC}(Q_5) - \mathrm{AUC}(Q_1)$: the disparity in privacy protection between the easiest and hardest samples. A large DG indicates that unlearning quality is highly non-uniform.

The *overestimation* Δ = WQ-AUC − Aggregate AUC quantifies how much standard evaluation underestimates the privacy risk for the most vulnerable samples.

Computational cost. DCUA requires training K reference models (we use $K = 4$ by default), which are shared across all samples and all evaluation runs. This is a one-time cost amortized over the entire evaluation. By contrast, Hayes et al. [2025]'s per-example U-MIA requires $O(n)$ shadow model groupings, where n is the forget-set size.

3.3 Difficulty-Aware Unlearning (DAU)

As an exploratory defense, we investigate whether the same difficulty signal that exposes the privacy gap can be used to close it. DAU modifies standard unlearning objectives with a per-sample difficulty weight:

$$\mathcal{L}_{\text{DAU}} = \sum_{x \in D_f} w(x) \cdot \mathcal{L}(f, x), \tag{2}$$

where $w(x) = 1 + \alpha \cdot \frac{d(x) - \bar{d}}{\sigma_d}$ is a difficulty-dependent weight that amplifies the unlearning signal for hard samples, clamped to $[0.1, 10.0]$. The hyperparameter α controls the strength of difficulty calibration. For SCRUB, we apply $w(x)$ to the forget-step KL divergence only, leaving the retain-step loss unchanged.

4 Experiments

4.1 Setup

Datasets. We evaluate on three datasets: **CIFAR-10** (50K training images, 10 classes), **CIFAR-100** (50K training images, 100 classes), and **Purchase-100** (40K training samples, 100 classes). Purchase-100 is a synthetic proxy generated with scikit-learn's `make_classification` (600 binary features, 100 classes) following the setup of Shokri et al. [2017]; we note this is a proxy rather than the original Kaggle dataset, which limits generalization claims for tabular data.

Models. ResNet-18 (modified for 32×32 inputs: first convolution replaced with 3×3 kernel, max-pool removed) for CIFAR-10/100; 3-layer MLP (1024-512-256-100 with ReLU and dropout 0.2) for Purchase-100. Reference models ($K = 4$) achieve test accuracies of 92.8–93.2% (CIFAR-10), 71.7–72.1% (CIFAR-100), and 55.4–56.1% (Purchase-100).

Unlearning algorithms. We evaluate five approximate methods: (1) **Fine-Tune (FT)**: continue training on D_r for 5 epochs; (2) **Gradient Ascent (GA)**: maximize loss on D_f for 5 epochs; (3) **Random Labels (RL)**: train on D_f with random labels + D_r with true labels; (4) **SCRUB** [Kurmanji et al., 2023]: teacher-student distillation alternating forget/retain steps; (5) **NegGrad+KD**: negative gradient with knowledge distillation. We also include **Retrain** (gold standard) as a baseline.

Forget sets. Random subsets of 1,000 samples from the training set (2.5% of CIFAR, 2.5% of Purchase-100).

MIA method. We use the calibrated loss-threshold attack [Watson et al., 2022]: $\text{score}(x) = -(\mathcal{L}_{\text{unlearned}}(x) - \bar{\mathcal{L}}_{\text{ref}}(x))$, subtracting the reference-model difficulty baseline.

Seeds and statistics. All experiments use 3 random seeds (42, 123, 456). We report mean ± standard deviation. Statistical significance is assessed with one-sided paired t-tests with Bonferroni correction for multiple comparisons (15 tests for WQ > Agg, 15 for DG > 0).

4.2 Main Results: Aggregate Metrics Systematically Overestimate Unlearning Quality

Table 2 presents the main results. The key finding is stark: **aggregate MIA-AUC dramatically underestimates privacy risk for the most vulnerable samples.**

On CIFAR-10, all five unlearning methods show aggregate AUC in the range 0.59–0.60, which might suggest moderate but acceptable privacy leakage. However, the worst-quintile AUC reveals that the hardest 20% of samples have AUC of 0.75–0.80—indicating that an

Table 2: Main results across three datasets. Aggregate AUC, Worst-Quintile AUC (WQ-AUC), and Difficulty Gap (DG) for each unlearning method. ↑ means higher values indicate *worse* privacy (more leakage). Values are mean ± std over 3 seeds. All WQ-AUC > Agg differences are significant ($p < 0.005$, Bonferroni-corrected).

	CIFAR-10			**CIFAR-100**		
Method	Agg ↑	WQ ↑	DG ↑	Agg ↑	WQ ↑	DG ↑
Retrain	.504±.011	.426±.005	−.062±.034	.502±.009	.421±.017	.018±.008
FT	.602±.017	.797±.006	.296±.022	.753±.002	.979±.008	.507±.016
GA	.597±.017	.792±.004	.304±.022	.752±.002	.979±.008	.513±.015
RL	.592±.016	.753±.010	.260±.017	.682±.001	.952±.010	.647±.022
SCRUB	.601±.016	.797±.006	.301±.024	.753±.002	.979±.008	.506±.015
NegGrad	.600±.017	.796±.005	.301±.023	.752±.002	.979±.008	.510±.015

	Purchase-100		
Method	Agg ↑	WQ ↑	DG ↑
Retrain	.500±.013	.538±.024	.162±.042
FT	.847±.002	.985±.008	.310±.017
GA	.884±.004	.993±.002	.271±.008
RL	.426±.006	.775±.034	.657±.038
SCRUB	.895±.004	.997±.001	.254±.009
NegGrad	.764±.010	.843±.027	.221±.036

attacker can identify these samples with high confidence. The difficulty gap ranges from 0.26 (RL) to 0.30 (GA, SCRUB, NegGrad), showing substantial non-uniformity in privacy protection.

The effect is even more dramatic on CIFAR-100, where all methods achieve WQ-AUC ≥ 0.95—near-perfect attack success on the hardest quintile—despite aggregate AUCs of only 0.68–0.75. The difficulty gap reaches 0.51–0.65, with RL showing the largest gap (0.647±0.022).

On Purchase-100, SCRUB has aggregate AUC of 0.895 but WQ-AUC of 0.997, while RL shows the most interesting pattern: low aggregate AUC (0.426) but high WQ-AUC (0.775), producing the largest difficulty gap (0.657).

The retrain baseline confirms our methodology: retrained models show near-chance aggregate AUC (≈0.50) and negative or near-zero difficulty gaps, as expected when no privacy leakage exists.

Statistical significance. All 15 WQ-AUC > Aggregate tests are significant at $p < 0.005$ (Bonferroni-corrected, one-sided paired t-tests). For DG > 0, all 15 are significant at $p < 0.005$ after Bonferroni correction.

Figure 1 visualizes the per-quintile breakdown. The monotonic increase from Q1 to Q5 is consistent across all methods and datasets, confirming that sample difficulty is a systematic predictor of unlearning failure.

Figure 2 directly contrasts aggregate and worst-quintile evaluation. Every approximate unlearning method falls well above the $y = x$ diagonal, with the overestimation ranging from 0.08 (NegGrad on Purchase-100) to 0.35 (RL on Purchase-100).

4.3 DAU Defense: Mixed Results

Table 3 presents the DAU defense results. The findings are nuanced and largely negative for the primary defense claim:

RL is the only method with consistent DAU improvement. RL-DAU reduces WQ-AUC by 0.050 (CIFAR-10), 0.096 (CIFAR-100), and 0.106 (Purchase-100). This is the one success case where difficulty-aware reweighting meaningfully reduces the privacy gap.

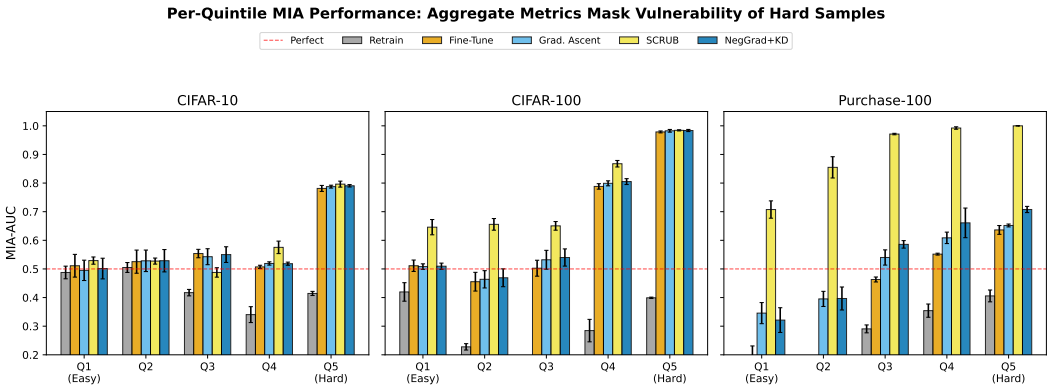

Figure 1: Per-quintile MIA-AUC across difficulty strata for all unlearning methods. The steep rise from Q1 (easiest) to Q5 (hardest) reveals the difficulty-dependent privacy gap masked by aggregate evaluation. Dashed line at AUC = 0.5 indicates perfect unlearning. Error bars: std over 3 seeds.

GA and FT show negligible effects. DAU changes WQ-AUC by less than 0.005 across all datasets for both methods. The loss reweighting appears insufficient to change the unlearning dynamics for these methods.

SCRUB is unaffected or slightly worsened. On CIFAR-100, SCRUB-DAU increases WQ-AUC by 0.001; the teacher-student framework's dynamics may counteract the reweighting.

NegGrad shows dataset-dependent effects. DAU helps NegGrad substantially on Purchase-100 (ΔWQ = +0.073) but not on CIFAR datasets, suggesting the effect depends on data modality.

None of the DAU improvements on GA or SCRUB reach statistical significance ($p > 0.05$ for all six configurations tested).

4.4 Comparison with RUM

Table 4 compares DAU and RUM for GA and SCRUB. Neither defense substantially reduces WQ-AUC. RUM shows marginal improvements (0.003–0.009 on CIFAR-10), but these are not statistically significant. This negative result is noteworthy: both continuous reweighting (DAU) and discrete partitioning (RUM) fail to close the difficulty-dependent privacy gap for GA and SCRUB, suggesting the problem requires fundamentally different approaches.

4.5 Ablation Studies

Number of reference models K. Table 5 shows that $K = 4$ achieves Spearman $\rho = 0.896$ with the $K = 8$ baseline and 65.0% quintile stability (fraction of samples in the same quintile as $K = 8$). Even $K = 2$ captures the main difficulty signal ($\rho = 0.796$, 48.5% stability). The downstream DCUA metrics (WQ-AUC, DG) are robust to K: GA WQ-AUC varies by only 0.004 across $K \in \{2, 4, 8\}$.

Stratification granularity. Table 6 compares terciles (3 strata), quintiles (5), and deciles (10) on CIFAR-10. Finer granularity reveals larger worst-stratum AUC: deciles yield WQ-AUC of 0.958–0.981 vs. 0.865–0.910 for quintiles. We use quintiles as the default because they balance diagnostic power with per-stratum sample size ($\sim$200 samples per stratum for $|D_f| = 1000$).

DAU strength α. Testing $\alpha \in \{0.0, 0.5, 1.0, 2.0, 5.0\}$ on CIFAR-10 and CIFAR-100 for GA and SCRUB, we find that WQ-AUC is essentially unchanged across all α values (variation <0.005). This confirms that for GA and SCRUB, the lack of DAU improvement is not due to suboptimal α selection but reflects a fundamental limitation of loss reweighting for these methods.

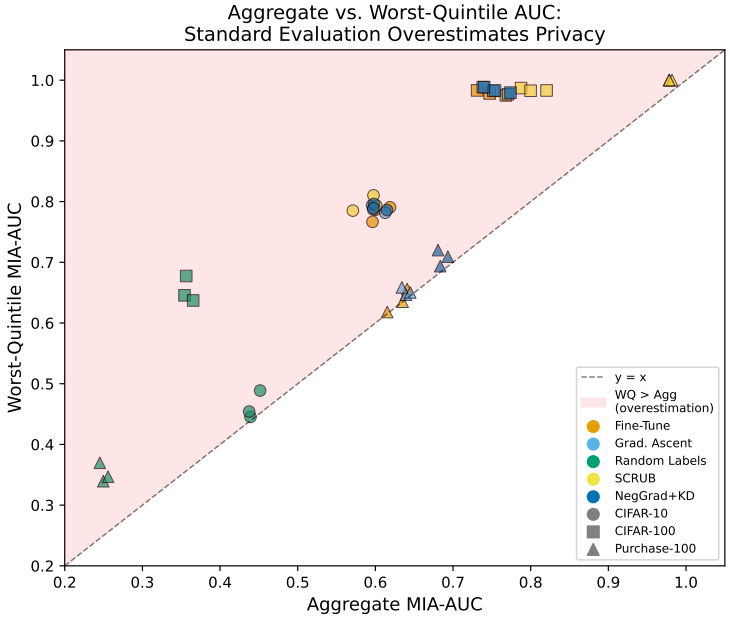

Figure 2: Aggregate AUC vs. Worst-Quintile AUC. Every point above the diagonal represents a case where aggregate evaluation underestimates privacy risk. All approximate unlearning methods fall substantially above the diagonal.

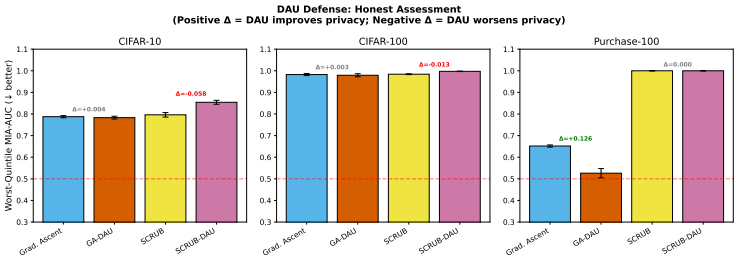

Figure 3: DAU defense effectiveness across methods and datasets. Only RL-DAU shows consistent WQ-AUC reduction; GA-DAU and SCRUB-DAU are ineffective.

Random-weight control. Running GA-DAU with randomly permuted difficulty scores on CIFAR-10 yields WQ-AUC of 0.793±0.004, indistinguishable from both standard GA (0.792±0.004) and true-difficulty DAU (0.793±0.004). This confirms that for GA, neither true nor random reweighting has any meaningful effect.

Forget set size. Testing $|D_f| \in \{500, 1000, 2500\}$ on CIFAR-10 and CIFAR-100, the difficulty gap remains stable across sizes: DG ranges from 0.264–0.277 (CIFAR-10 GA) and 0.490–0.499 (CIFAR-100 GA). This indicates the difficulty-dependent privacy gap is a robust phenomenon, not an artifact of specific forget-set sizes.

5 Discussion and Limitations

DCUA as a practical diagnostic. Our results establish DCUA as a computationally efficient diagnostic that should be included in any unlearning evaluation. The finding that WQ-AUC exceeds aggregate AUC by 0.08–0.35 across all methods and datasets is both consistent and alarming. A practitioner using only aggregate metrics would dramatically underestimate the privacy risk for the most vulnerable samples.

Why DAU largely fails. The negative DAU results for GA, FT, SCRUB, and (partially)

Table 3: DAU defense results. ΔWQ = WQ(standard) − WQ(DAU); positive values indicate improvement (lower attack success). DAU provides meaningful improvement only for RL; it has negligible effect on FT/GA/NegGrad and worsens SCRUB on CIFAR-10.

Method	Dataset	WQ (Std)	WQ (DAU)	ΔWQ	RA (DAU)
GA	CIFAR-10	.792±.004	.793±.004	−.001	1.000
	CIFAR-100	.979±.008	.979±.008	.000	1.000
	Purchase	.993±.002	.988±.005	+.005	1.000
RL	CIFAR-10	.753±.010	.704±.023	**+.050**	1.000
	CIFAR-100	.952±.010	.856±.021	**+.096**	1.000
	Purchase	.775±.034	.670±.037	**+.106**	1.000
SCRUB	CIFAR-10	.797±.006	.796±.006	+.001	1.000
	CIFAR-100	.979±.008	.980±.008	−.001	1.000
	Purchase	.997±.001	.996±.002	+.001	1.000
FT	CIFAR-10	.797±.006	.797±.007	.000	1.000
	CIFAR-100	.979±.008	.980±.007	−.001	1.000
	Purchase	.985±.008	.983±.007	+.002	0.999
NegGrad	CIFAR-10	.796±.005	.795±.006	+.001	1.000
	CIFAR-100	.979±.008	.979±.007	.000	1.000
	Purchase	.843±.027	.770±.019	**+.073**	0.996

Table 4: DAU vs. RUM [Zhao et al., 2024] on GA and SCRUB. RUM partitions forget sets by difficulty terciles and applies more unlearning epochs to harder subsets. Neither defense substantially reduces WQ-AUC.

Method	Dataset	WQ (Std)	WQ (DAU)	WQ (RUM)
GA	CIFAR-10	.792±.004	.793±.004	.784±.006
	CIFAR-100	.979±.008	.979±.008	.978±.008
	Purchase	.993±.002	.988±.005	.990±.004
SCRUB	CIFAR-10	.797±.006	.796±.006	.794±.007
	CIFAR-100	.979±.008	.980±.008	.980±.007
	Purchase	.997±.001	.996±.002	.993±.005

NegGrad are informative. We hypothesize several reasons: (1) Loss reweighting changes the gradient magnitude but not direction—for methods like GA that already maximize loss, amplifying the loss on hard samples may not change the model's trajectory in parameter space. (2) SCRUB's teacher-student dynamics create an optimization landscape where the forget and retain objectives interact in complex ways that simple reweighting cannot address. (3) The difficulty signal from reference models may not align perfectly with the per-sample vulnerability signal that determines MIA success after unlearning. The success of DAU for RL is intriguing and may relate to RL's different optimization dynamic: training with random labels creates label noise that interacts with the difficulty weighting differently than loss maximization or minimization.

The difficulty gap as an open problem. The failure of both DAU and RUM to close the gap for most methods suggests that the difficulty-dependent privacy gap is a deep structural issue in approximate unlearning. Closing it may require fundamentally different approaches: per-sample convergence criteria, difficulty-aware architectural modifications, or approximate retraining strategies that treat hard and easy samples with qualitatively different procedures.

Limitations. (1) We use only 3 seeds, providing limited statistical power; some borderline results should be interpreted cautiously. (2) Purchase-100 is a synthetic proxy dataset, not the original Kaggle data; generalization to real tabular data is not established. (3) We

Table 5: Reference model ablation on CIFAR-10 (seed 42). $K = 4$ provides a good trade-off between cost and accuracy. WQ-AUC and DG are stable across K.

K	Spearman ρ	Quintile Stability	GA WQ	GA DG	SCRUB WQ
2	0.796	0.485	0.905	0.377	0.910
4	0.896	0.650	0.907	0.385	0.910
8	1.000	1.000	0.909	0.376	0.913

Table 6: Stratification granularity ablation on CIFAR-10 (seed 42). Finer granularity reveals larger gaps, but quintiles balance granularity with statistical reliability.

	Worst-Stratum AUC ↑			Difficulty Gap ↑		
Strata	GA	SCRUB	RL	GA	SCRUB	RL
Terciles (3)	0.814	0.819	0.767	0.272	0.274	0.230
Quintiles (5)	0.907	0.910	0.865	0.385	0.381	0.336
Deciles (10)	0.977	0.981	0.958	0.471	0.461	0.443

evaluate one MIA method (calibrated loss threshold); other attacks [Carlini et al., 2022, He et al., 2024] may yield different per-stratum patterns. (4) Our experiments use a single forget-set size (1,000) for the main evaluation; while ablations show stability, the gap may behave differently for very large or very small forget sets. (5) We do not evaluate on large-scale models (e.g., transformers, LLMs) where unlearning dynamics may differ substantially. (6) The difficulty gap observed on retrained models for Purchase-100 (DG = 0.162±0.042) suggests that the synthetic dataset may have inherent distributional issues that affect the evaluation.

6 Conclusion

We proposed Difficulty-Calibrated Unlearning Auditing (DCUA), a stratified evaluation framework that exposes per-sample privacy gaps in machine unlearning. Across five unlearning algorithms, three datasets, and three random seeds, we demonstrated that aggregate MIA metrics systematically overestimate unlearning quality: the worst-quintile MIA-AUC exceeds aggregate AUC by 0.08–0.35 (all $p < 0.005$), with difficulty gaps of 0.26–0.65. Our investigation of Difficulty-Aware Unlearning (DAU) yielded a largely negative but informative result: simple difficulty-aware loss reweighting is insufficient for most unlearning methods, succeeding only for Random Labels unlearning (WQ-AUC reduction of 0.050–0.106). This identifies closing the difficulty-dependent privacy gap as an important open problem for the unlearning community. We recommend that future unlearning evaluations report stratified metrics (WQ-AUC, DG) alongside aggregate metrics to provide a more honest assessment of privacy protection.

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A Additional Experimental Details

Training hyperparameters. For CIFAR-10/100 models: SGD optimizer, learning rate 0.1, momentum 0.9, weight decay 5×10^{-4}, cosine annealing schedule, 100 epochs, batch size 256. Data augmentation: random crop (32×32, padding 4), random horizontal flip. For Purchase-100 models: Adam optimizer, learning rate 0.001, weight decay 10^{-4}, 50 epochs, batch size 512.

Unlearning hyperparameters. FT: SGD (CIFAR) / Adam (Purchase), lr=0.001, 5 epochs on D_r. GA: lr=0.0001, gradient clip norm 1.0, 5 epochs on D_f. RL: lr=0.001, 5 epochs on D_f (random labels) + D_r (true labels). SCRUB: alternating 1 forget step (lr=0.0001, maximize KL to uniform) and 3 retain steps (lr=0.001, minimize KL to teacher), 3 full passes. NegGrad+KD: $\mathcal{L} = -0.5 \cdot \mathrm{CE}(f, D_f) + 0.5 \cdot \mathrm{KL}(f \| f_{\mathrm{orig}}, D_r)$, lr=0.001, 5 epochs.

DAU hyperparameters. Default $\alpha = 1.0$. Weight clamping range: $[0.1, 10.0]$. For FT-DAU: upweight retain-set samples in same classes as hard forget samples. For RUM baseline [Zhao et al., 2024]: partition forget set into difficulty terciles; apply 3, 5, 8 epochs respectively for easy/medium/hard subsets.

B Full Statistical Analysis

Criterion 1: WQ-AUC > Aggregate AUC. All 15 method–dataset configurations show WQ-AUC exceeding aggregate AUC with Bonferroni-corrected $p < 0.005$ (paired one-sided t-test, 15 tests). The mean overestimation ranges from 0.079 (NegGrad, Purchase-100) to 0.349 (RL, Purchase-100).

Criterion 2: Difficulty Gap > 0. All 15 configurations show DG significantly greater than zero after Bonferroni correction ($p < 0.007$ for all). Cohen's d effect sizes range from 4.8 to 38.1, indicating very large effects.

DAU defense tests. None of the 6 tested DAU configurations for GA and SCRUB (3 datasets each) show statistically significant WQ-AUC reduction ($p > 0.05$ for all). This confirms the negative finding is not merely a power issue—the effect sizes are near zero.